

# Benchmark Analysis of Jetson TX2, Jetson Nano and Raspberry PI using Deep-CNN

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**Abstract**—Hardware, low power consumption, high accuracy and performance are crucial factors for deep learning applications. High level graphics processing units (GPU) are commonly used in high performance deep learning applications. However, it is a lot in terms of cost and power consumption to build a high-performance platform. In this study, performances of single-board computers in NVIDIA Jetson Nano, NVIDIA Jetson TX2 and Raspberry PI4 through CNN algorithm created by using fashion product images dataset are compared. 2D CNN model has been developed so as to classify 13 different fashion products in tests. Data set is comprised of 45K pictures. Parameters for performance analysis has been defined as consumption (GPU, CPU, RAM, Power), accuracy and cost. Data set is divided into parts of 5K, 10K, 20K, 30K and 45K in training and test of the model in order to expand on the differences of single-board computers. Eventually, performance of the embedded system boards in different data set in CNN algorithm is analyzed. It is, thus, aimed to attain high accuracy preference by minimum hardware requirements in deep learning applications.

**Keywords**—Benchmark, Convolutional Neural Network, Jetson TX2, Jetson Nano, Raspberry PI

## I. INTRODUCTION

Computers have become hardware that we can operate almost unlimitedly with its continuous development from the time of its invention until up-to-date. Computers that are large enough to cover a room when it first comes out offer opportunities to perform different applications by increasing their processing capacity much more and by becoming able to fit in the palm of its hand. In some applications, instead of high-cost and physically heavy computers; small, portable, low-cost, and high-performance computers with relatively high features are desired to bring solutions. In such cases, single-board computers are preferred. Single-board computers called that are built on a single circuit board and have microprocessors, memory, input/output, and other functional features [1]. Single-board computer systems with high applicability and low cost have become frequently preferred up-to-date, as it can be combined with other technology fields [2]. The technological development of single-board computers and their approach to the performance of personal computers have strengthened its use in the field of artificial intelligence. Likewise, the fact that some single-board computers host both CPU and GPU have increased the applicability of deep learning, which is one of the sub-branches of artificial intelligence, in single-board computers. Single-board computers are one of the most important options for deep learning applications where the goal is autonomous/mobile systems [3].

Deep learning is a technique that makes use of many nonlinear hidden layers for supervised/unsupervised feature

extraction, transformation, pattern analysis, and classification, where each successive layer takes the output from the previous layer as input [4]. Deep learning has architectures such as Convolutional Neural Network (CNN), Recurrent Neural Network (RNN), Long-Short Term Memory (LSTM), Restricted Boltzmann Machines (RBM), deep auto-encoder [5]. Deep learning is used in areas such as face recognition, voice recognition, fingerprint readers, license plate readers, driverless cars, and security cameras [6]. From deep learning architectures, CNN algorithms are implemented in many different areas, mainly in the field of image and audio processing, such as natural language processing (NLP), biomedical. It has very high results are achieved, especially in image processing [7].

In this study, a 2D-CNN model was developed to measure the performance of deep learning applications on a single-board computer. The Model classifies fashion products by 13 different fashion types. The model has been trained and tested in the five different size datasets on single-board computers (NVIDIA Jetson Nano, NVIDIA Jetson TX2, and Raspberry PI 4). For the evaluation of performance differences in single-board computers, the data set was clustering in the training and testing of the model. The study aimed to contribute to high accuracy preference with minimum hardware in deep learning applications for image processing on single-board computers.

## II. RELATED WORK

In 2012, interest in deep learning increase with revival, as Krizhevsky vd. (2012)'s ImageNet Large Scale Visual Recognition Competition (ILSVRC) showed significantly higher image classification accuracy [8]. Due to the increased usefulness of deep learning, several studies have been conducted to effectively use deep learning techniques on single-board computers / embedded platforms. In the researches, while Raspberry Pi provides low power and energy saving performance, it is concluded that NVIDIA Jetson platforms such as Jetson Nano, Jetson TX1, Jetson TX2 have higher performance because they have higher speed GPUs. Compared to Jetson and Raspberry Pi performances, Jetson provides higher performance [9]. For this reason, NVIDIA Jetson series leads in the field of single-board computing in deep learning applications since it offers developer kits with various features. Some of the features of the Jetson series are energy efficiency, light and portable, making it easy to use and high performance per watt. NVIDIA's Jetson series is seen as the most widely used accelerator in the machine learning extraction phase. Compared to Jetson TX2 and Jetson Nano in CNN tests, Jetson Nano shows 2 times lower performance [10]. Jo et al. (2020) as a result of their studies, Jetson Nano developer kit has been evaluated to be higher in some deep learning applications than the Raspberry Pi kit [10]. When comparing

the performance of Jetson Nano and Jetson TX2 developers with OpenCV using the OpenCv Template Matching method, it appears that the Jetson TX2 developer kit is on average three times faster than the Jetson Nano developer kit. However, both are very low in terms of processing time [11]. Peng et. al., as a result of customized study on GPU systems, Jetson TX2 has been shown to have the best power efficiency compared to other Jetson developer kits [12]. In the performance analysis of Kang et al. On CNN architectures, they observed that Jetson TX2 developer achieved the highest efficiency in the Jetson series [13]. Behavior knowledge of some developers for parallel computing was evaluated on Jetson TX1 and Raspberry Pi development board. The quad core CPU and CUDA compatible GPU, Jetson TX1 development board has been observed to be more effective at high performance angles with low energy consumption. It has been stated that Raspberry Pi 2 is not very suitable for high performance processes [14].

In Bordignon and Wangenheim (2019)' study have developed a benchmark using NVIDIA Jetson TX2 to assist in deciding which architecture is more suitable for the current hardware to measure how well the Jetson platform behaves in real world conditions and to see its results. Microsoft COCO (80 different object types and 40670 images) general-purpose dataset and KITTI Stereo Vision (400 images of pedestrian and road scenarios under common conditions) were used for testing. The GPU temperature values and memory usage measured in these datasets are given graphically. SSD V1 application over MobileNets base network is stated to be the best performing model in Jetson TX2 under normal conditions in both data sets [15]. In a study using Nvidia Jetson TX2 and Movidius single- board computers (embedded platform), the performance of various applications of YOLO, SSD, RCNN, R-FCN and SqueezeDet was evaluated using an internally proprietary data set consisting of more than 10 thousand closed images. The evaluations were made on the CPU and GPU based on the continuous extraction approach with the total extraction delay of a single test image and repetitive camera shooting. They confirmed that the GPU is much faster than a CPU in matrix arithmetic such as convolution due to its high bandwidth and parallel computing capabilities. Experimental results have shown that Tiny YOLO-416 and SSD (VGG-300) are the fastest and Faster-RCNN (Initial ResNet-v2) and R-FCN (ResNet-101) have high detection accuracy. YOLO v3-416 has been reported to yield relatively accurate results within a reasonable time for person detection on embedded platforms [16].

### III. SINGLE-BOARD COMPUTERS

Unlike desktop or personal computers, single-board computer systems are a system with high performance in its simple architecture. Although single-board computers can't perform many functions that a personal computer can perform, the difference in design and construction reveals its intended use [17]. There are many single-board computer systems that offer possibilities to develop both hardware and software and include CPU / GPU. Jetson Nano / TX2, Raspberry Pi, BeagleBoard, Asus Tinker Board are some of them. Single- board computers are used in ATMs, medical diagnostics, precision agriculture, smart home systems, robotic systems and many other fields [18].

Raspberry Pi is a single credit board-sized PC printed circuit board that can be used for many things your computer does, such as games, word processing, spreadsheets, and also

for playing HD video [19]. The Raspberry Pi 4 Model B is the latest in the popular Raspberry Pi computer series. While maintaining backward compatibility and similar power consumption, compared to the previous generation Raspberry Pi 3 Model B+, it offers significant increases in processor speed, multimedia performance, memory and connectivity. Raspberry Pi 4 Model B provides desktop performance comparable to entry-level x86 personal computer (PC) systems [20].

NVIDIA Jetson Nano, the entry-level board of the NVIDIA Jetson ecosystem, is a small, powerful single- board computer that allows parallel operation of multiple neural networks for applications such as image classification, object detection, segmentation, and speech processing. It has a comprehensive development environment (JetPack SDK) and libraries developed for embedded applications, deep learning, IoT, computer vision, graphics, multimedia and more. Using Jetson Nano with a GeForce-enabled graphics processor (GPU) using the same CUDA cores creates a very powerful development environment for applications [21]. In addition, Jetson Nano has a CPU-GPU heterogeneous architecture in which the operating system can be booted by the CPU and can be programmed to speed up complex machine learning tasks of the CUDA capable GPU [13]. Artificial intelligence reveals low power consumption in running algorithms. The medium-sized board of the Nvidia Jetson ecosystem, Jetson TX2, is higher than Jetson Nano. It is very useful for computer vision and deep learning. Table 1 gives the technical specifications of Raspberry Pi4, Jetson Nano and Jetson TX2 boards.

TABLE I. TECHNICAL SPECIFICATIONS OF JETSON NANO, TX2 AND RASPBERRY PI4 [9,13,22]

|                         | <b>Rasph. PI 4</b>                        | <b>Jetson Nano</b>                         | <b>Jetson TX2</b>                                                 |
|-------------------------|-------------------------------------------|--------------------------------------------|-------------------------------------------------------------------|
| <b>Performance</b>      | 13.5 GFLOPS                               | 472 GFLOPS                                 | 1.3 TFLOPS                                                        |
| <b>CPU</b>              | Quad-core ARM Cortex-A72 64-bit @ 1.5 GHz | Quad-Core ARM Cortex-A57 64-bit @ 1.42 GHz | Quad-Core ARM Cortex-A57 @ 2GHz + Dual-Core NVIDIA Denver2 @ 2GHz |
| <b>GPU</b>              | Broadcom Video Core VI (32-bit)           | NVIDIA Maxwell w/ 128 CUDA cores @ 921 MHz | NVIDIA Pascal 256 CUDA cores @ 1300MHz                            |
| <b>Memory</b>           | 8 GB LPDDR4                               | 4 GB LPDDR4 @ 1600MHz, 25.6 GB/s           | 8GB 128-bit LPDDR4 @ 1866Mhz, 59.7 GB/s                           |
| <b>Networking</b>       | Gigabit Ethernet / Wi-Fi 802.11ac         | Gigabit Ethernet / M.2 Key E               | Gigabit Ethernet, 802.11ac WLAN                                   |
| <b>Display</b>          | 2x micro-HDMI (up to 4Kp60)               | HDMI 2.0 and eDP 1.4                       | 2x DSI, 2x DP 1.2 / HDMI 2.0 / eDP 1.4                            |
| <b>USB</b>              | 2x USB 3.0, 2x USB 2.0                    | 4x USB 3.0, USB 2.0 Micro-B                | USB 3.0 + USB 2.0                                                 |
| <b>Other</b>            | 40-pin GPIO                               | 40-pin GPIO                                | 40-pin GPIO                                                       |
| <b>Video Encode</b>     | H.264(1080p30)                            | H.264/H.265 (4Kp30)                        | H.264/H.265(4Kp60)                                                |
| <b>Video Decode</b>     | H.265(4Kp60), H.264(1080p60)              | H.264/H.265 (4Kp60, 2x 4Kp30)              | H.264/H.265 (4Kp60)                                               |
| <b>Camera</b>           | MIPI CSI port                             | MIPI CSI port                              | MIPI CSI port                                                     |
| <b>Storage</b>          | Micro-SD                                  | 16 GB eMMC                                 | 32GB eMMC                                                         |
| <b>Power under load</b> | 2.56W-7.30W                               | 5W-10W                                     | 7.5W-15W                                                          |
| <b>Price</b>            | \$35                                      | \$89                                       | \$399                                                             |

#### IV. EXPERIMENTAL STUDY

In the study benchmarked experimental tests on the classification of traditional 2D images on Jetson TX2, Jetson Nano, and Raspberry Pi. Classification of 2D images has provided using the TensorFlow library with Spyder software on both GPU and CPU.

##### A. Data Set Preparation

A DeepFashion2 dataset was used in the CNN model developed for performance tests. DeepFashion2 includes 13 popular clothing categories and 45K images collected from both department stores and consumers [23] (Table 2).

TABLE II. FASHION IMAGES TYPES OF DEEPPHOTO2

| category_id | category_name                   |
|-------------|---------------------------------|
| 1           | represents short sleeve top     |
| 2           | represents long sleeve top      |
| 3           | represents short sleeve outwear |
| 4           | represents long sleeve outwear  |
| 5           | represents vest                 |
| 6           | represents sling                |
| 7           | represents shorts               |
| 8           | represents trousers             |
| 9           | represents skirt                |
| 10          | represents short sleeve dress   |
| 11          | represents long sleeve dress    |
| 12          | represents vest dress           |
| 13          | represents sling dress          |

The dataset has clustered into 5K, 10K, 20K, 30K, and 45K to view the performance ranges of the model to be trained on CPU and GPU. The K-Means algorithm was used to cluster the data into groups based on similar numeric attributes [24]. The value  $k=21$  is determining the number of clusters in k-Means.  $K = 21$ , randomly selects 21 center points in the

algorithm. By calculating the distance between each data and its randomly determined center points, it assigns the data to a set relative to the nearest center point [25]. After clustering was completed, 21 different clusters of 5K sizes were formed. 21 clusters were combined as 1,2,4,6 and 9 into groups to obtain 5 different data sets to be used in the training of the model. Thus, the goal data sets have been obtained. In the final step, each dataset is divided into training and test data. Here, the datasets are divided into 70% training and 30% testing.

##### B. Development of CNN Model

Convolutional neural network (CNN) is a deep-feed artificial neural network algorithm commonly used in computer vision problems [26]. The CNN model in which the experimental comparison will be made is modeled in Figure 1. The CNN model is implemented step by step in the following architecture.

- Input:  $32 \times 32$
- Conv5:  $5 \times 5$  size, 32 filters
- ReLU:  $\max(0, h\theta(x))$
- Pool:  $2 \times 2$  size
- Conv5:  $5 \times 5$  size, 64 filters
- ReLU:  $\max(0, h\theta(x))$
- Pool:  $2 \times 2$  size
- FC: 1024 Hidden Neurons
- Dropout:  $p = 0.2$
- FC: 13 Softmax Output Class

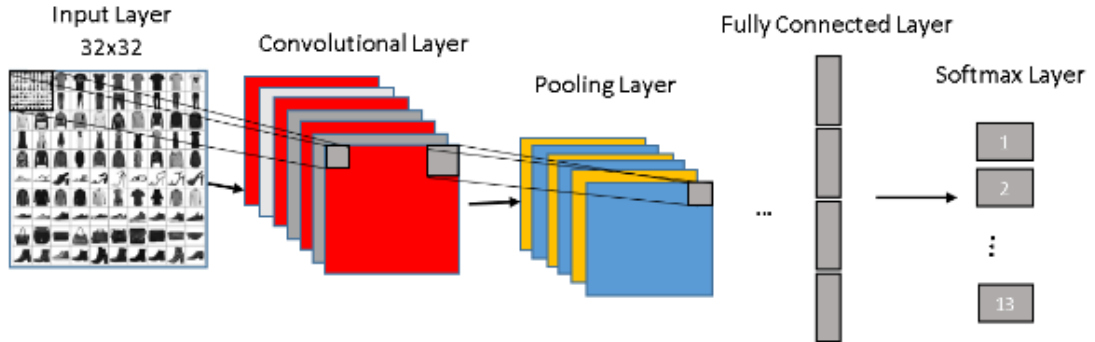


Fig. 1. CNN classification model

##### C. Metric

Accuracy (Equation 1) that commonly used in classification models is preferred for comparing the performance of models [5].

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

Where TP = True Positives, TN = True Negatives, FP = False Positives, and FN = False Negatives.

#### V. RESULTS AND DISCUSSION

This section discusses the performance comparison of the CNN classification model with data sets of different sizes in the Jetson and Raspberry PI architectures. The Deep Neural Network library (cuDNN) library was used for CNN model testing. cuDNN has been favored for providing high levels of applications for forward and backward layers of convolution, pooling, normalization, and activation. Benchmarks were carried out on the CPU in Raspberry Pi and on the GPU in Jetson boards. Training and test performances of the developed model were conducted separately for data sets of 5 different sizes. JTOP software preferred to benchmark

performance differences. The Jtop utility has been used for visualizes CPU, GPU, memory, and power status [27].

The results of the CNN model's performance tests on single-board computers developed for fashion dataset classification are given in Table 3. Parameters of in these performance tests, it has determined as accuracy (%), time (sec), memory(GB), CPU(W), and GPU (W). Idle measurements of the boards were made before beginning the tests. The Jetson TX2 has been shown to have higher source and power consumption due to its high hardware capabilities. When the classification of the model is analyzed by time, it is seen that it takes place in a faster time over GPU (Jetson TX2 and Jetson Nano) and increases in time with the increase in data set. It reaches very high times in the classification made via CPU (Raspberry PI). Memory errors occur in Jetson Nano

and Raspberry PI when the dataset size is 20K. The Jetson TX2 has had a maximum memory consumption of 45K datasets. The increase in power consumption increases the power consumption in both CPU and GPU. It has been observed that power consumption is more stable with big data increase in Jetson TX.

In single-board computers, epoch=100 was determined optimally for the models to achieve maximum accuracy performance. The positive effect of the size of the dataset on accuracy in deep learning applications was reflected in the tests conducted. The model has the highest accuracy with 97.8% with 45k dataset on the Jetson TX2. Single-board computers have been achieving very approximate accuracy to each other in the 5K dataset.

TABLE III. BENCHMARKING JETSON AND RASPBERRY PI

|         | Acc (%) |      |      | Time(sec) |      |     | Memory (GB) |      |     | CPU(Power/W) |      |      | GPU(Power/W) |      |    |
|---------|---------|------|------|-----------|------|-----|-------------|------|-----|--------------|------|------|--------------|------|----|
| Dataset | TX2     | Nano | PI   | TX2       | Nano | PI  | TX2         | Nano | PI  | TX2          | Nano | PI   | TX2          | Nano | PI |
| Idle    | -       | -    | -    | -         | -    | -   | 1,9         | 1,5  | 1,4 | 0,675        | 0,47 | 0,30 | 2,6          | 0,76 | -  |
| 5K      | 87,6    | 87,5 | 87,2 | 23        | 32   | 173 | 2,6         | 2,0  | 2,1 | 2,23         | 1,50 | 3,5  | 5,27         | 2,23 | -  |
| 10K     | 93,8    | 93,9 | 91,6 | 32        | 58   | 372 | 3,1         | 2,75 | 2,6 | 2,78         | 2,32 | 3,6  | 5,32         | 3,25 | -  |
| 20K     | 94,6    | 94,5 | -    | 52        | -    | 462 | 4,5         | ERR  | 4,0 | 3,76         | -    | 3,9  | 5,22         | -    | -  |
| 30K     | 96,4    | -    | -    | 122       | -    | -   | 5,2         | ERR  | ERR | 4,25         | -    | -    | 5,74         | -    | -  |
| 45K     | 97,8    | -    | -    | 235       | -    | -   | 6,5         | ERR  | ERR | 4,92         | -    | -    | 6,29         | -    | -  |

ERR: Memory Error, PI: Raspberry PI

The cost benchmark of single-board computers after performance tests is shown in Table 4. The cost of the Jetson TX2, which has high hardware capabilities, is higher than the others. Although Raspberry PI without NVIDIA GPU support is the most cost-effective hardware, it is not the right prefer for deep learning applications.

TABLE IV. PRICE COMPARISON JETSON AND PI

| Single-Board Computer | Price (\$) |
|-----------------------|------------|
| Raspberry PI          | 35         |
| Jetson Nano           | 89         |
| Jetson TX2            | 399        |

In the light of the benchmark, the choice of computers with high hardware characteristics provides an advantage in accuracy and performance in uses of deep learning algorithms. The most important disadvantage of highly equipped computers is that they are much higher in terms of cost. The use of the GPU in training large data sets provides a high time advantage.

## VI. CONCLUSION

In this study, performance tests of deep learning applications that have big data were conducted in Jetson TX2, Jetson Nano, and Raspberry PI boards. Performance tests had been covering accuracy, time, memory, and power consumption. As a result of the evaluation of performance tests, it is aimed to provide minimum hardware and cost and hardware selection in deep learning applications with big data. In this context, the model was developed with the CNN algorithm from deep learning algorithms. The DeepFashion2 dataset was preferred for the training and testing of the model. This dataset is clustered in 5 different sizes for detailing

performance tests. Training and testing were carried out on each hardware for 5 different clusters.

When the study results are evaluated, the Jetson TX2 has more power consumption, but it performs better in a shorter time, higher accuracy, and big data sets than others. In the last part, although Jetson TX2 has the highest in cost, low cost is not possible with high hardware features.

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